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The Illusion of Equivalence: Systematic FP16 Divergence in KV-Cached Autoregressive Inference

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ABSTRACT KV caching is deployed in virtually every production autoregressive transformer inference system, under an assumption of numerical equivalence to cache-free computation that is rarely tested in practice. We show this assumption fails under standard FP16 inference and, unlike prior nondeterminism studies that characterize divergence as an emergent statistical property of system configuration, localize it to a specific locus – the KV cache state, confirmed both by elimination (residual-stream patching) and by direct positive confirmation (KV cache tensor patching) – and to an architectural amplification mechanism, the Grouped-Query Attention (GQA) sharing ratio, that has not previously been connected to numerical divergence. Across three open-weight models (LLaMA-2-7B, Mistral-7B-v0.3, Gemma-2-2B) evaluated on GSM8K, we find cache-ON and cache-OFF execution paths produce deterministically divergent token sequences in 100% of runs, including under greedy decoding, ruling out sampling as a cause. Under our evaluation protocol (128-token truncation, few-shot formatting), cache-ON yields higher accuracy in 8 of 9 conditions; this difference serves as an indicator of systematic execution-path bias, not a measure of absolute reasoning ability, and does not affect the primary finding. We isolate the numerical mechanism via a controlled FP32 falsification (the flip rate drops to exactly 0.0%, and KL divergence decreases by more than eight orders of magnitude, eliminating the divergence entirely and isolating FP16 non-associativity as the cause) and a teacher-forced conditional divergence measurement that quantifies how much of the naively observed effect is pure numerical perturbation versus compounding context drift once trajectories first disagree. Layer-wise drift profiling reveals architecturally predictable propagation patterns tied to attention structure (Grouped-Query Attention vs. Multi-Head Attention) and head dimension. Activation patching of the entire residual stream fails to recover the cache-free trajectory, localizing the divergence to the KV cache state by elimination; direct patching of the KV cache tensors (K and V jointly, at decode step 0) then converts this into positive confirmation and reveals that a layer’s causal importance to correcting the divergence can dissociate sharply from how much divergence it passively accumulates. A targeted mixed-precision ablation further shows that elevating precision within the attention computation alone is insufficient, indicating that the effect requires precision to be sustained across the full forward pass. We additionally show that the divergence persists under BF16 at comparable severity and that final-answer outcomes—not just intermediate tokens—are affected in 98.4% of diverging cases, meaning that the divergence changes what the model outputs, not just how it gets there. We provide concrete guidance for practitioners on detecting and reasoning about this divergence in deployed inference systems.

INDEX TERMS Floating point arithmetic, large language models, numerical analysis, inference algorithms, reproducibility, autoregressive processes

I. INTRODUCTION

Autoregressive transformer inference is almost always based on the Key-Value (KV) cache: instead of recomputing attention over the entire prefix at every decoding step, previously computed keys and values are stored and retrieved as needed. This technique, found in practically every production serving system [13], efficient attention kernel [7], [6], and prompt-caching API, is typically assumed to be numerically equivalent to cache-free inference. This assumption is rarely tested directly by system designers because KV caching is treated as a pure performance optimization rather than a numerically consequential design choice. We show that it fails under standard FP16 inference, quantify when and how severely it breaks down, and supply practitioners with diagnostic tools and concrete guidance for detecting this divergence in their own systems.

The cache-ON and cache-OFF execution paths use different kernel layouts and, as a direct consequence, different floating-point accumulation orders under the non-associativity of FP16 arithmetic. This is the mechanism behind a systematic numerical divergence in the KV tensors written during decoding. Unlike the stochastic variability demonstrated in previous work [28], [4], this divergence is deterministic and reproducible: for the same input, model, and hardware/software environment, the cache-ON and cache-OFF inference paths always produce different output tokens. This determinism is central to our characterization. The divergence is not statistical variance that could average out with more data or longer generation; it is a specific, reproducible property of FP16 cached inference, confirmed later by our controlled FP32 falsification (Section III), which eliminates it entirely rather than merely reducing it.

Prior work has studied adjacent but distinct problems. Numerical precision research [16] addresses training stability and model compression [9], [8], not cache-ON/cache-OFF equivalence at inference time. KV cache optimization work [13], [18], [2] treats caching as a memory/throughput problem and assumes numerical equivalence to cache-free computation without verifying it. Mechanistic interpretability methods [15], [23] intervene on the residual stream but have not considered the stateful KV cache itself as a locus of divergence, which we do via activation patching (Section III). Closest to our work, concurrent studies of LLM inference non-determinism [26] characterize FP16/BF16 non-associativity across batch size, GPU count, and hardware configuration. We isolate a structurally distinct comparison, cache-ON versus cache-OFF execution on identical hardware and configuration, and further localize the effect specifically to the KV cache state and characterize its amplification via the GQA sharing ratio, neither of which prior work addresses. Concretely, this means our findings and theirs are not competing explanations of the same phenomenon: a mitigation such as LayerCast, designed for cross-configuration nondeterminism, would not be expected to resolve cache-ON/cache-OFF divergence on fixed hardware and a fixed configuration, because our divergence has a different mechanism (execution-path accumulation order

rather than cross-configuration variance) and a different causal locus, which we confirm in Section III both by elimination (residual-stream patching) and by direct positive confirmation (KV cache tensor patching, Section V-F).

We provide a systematic empirical analysis of KV cache divergence across three open-weight language models (LLaMA-2-7B, Mistral-7B-v0.3, and Gemma-2-2B) evaluated on the GSM8K data set [5] under greedy, top- k , and top- p sampling. Our analysis proceeds through a causal chain of experiments. We first establish that cache-ON and cache-OFF consistently produce different outputs across all conditions, then characterize how the divergence propagates layer by layer through the network. We identify FP16 accumulation order as the root cause through a controlled FP32 falsification, resolve an apparent contradiction between flip index and KL divergence through decision boundary analysis, and localize the divergence to the KV cache state rather than the residual stream via activation patching. We additionally introduce a teacher-forced conditional divergence measurement that isolates the pure numerical effect from divergence compounded by trajectories subsequently conditioning on different token histories, and report a targeted mixed-precision ablation, a BF16 reproduction, and a final-answer reconvergence analysis that together clarify the practical significance and boundaries of the effect.

Our results are consistent across all three models and all three sampling strategies, and point to a single mechanistic interpretation: FP16 rounding errors are written into KV tensors at each decode step, accumulate across layers and sequence positions, and cannot be corrected by downstream residual stream interventions. Architectural choices amplify this effect, e.g., grouped-query attention (GQA [2]) broadcasts a single FP16 rounding error to all attending query heads simultaneously, producing correlated rather than independent errors across heads and amplifying divergence relative to standard multi-head attention (MHA [21]).

In summary, our findings in this work include: **(i) Behavioral characterization.** For all scenarios, even with greedy decoding (which removes sampling as a possible explanation entirely), token divergence occurs in 100% of cases. Cache-ON is more accurate in 8 of 9 conditions (McNemar, Bonferroni-corrected $p < 0.05$). An unbiased perturbation would be expected to help and hurt accuracy in roughly equal measure across conditions; instead cache-ON wins in 8 of 9, which is the signature of a systematic execution-path bias. **(ii) Mechanistic root cause.** Switching to FP32 reduces KL divergence by over eight orders of magnitude, to a residual value of 2.5×10^{-8} attributable to floating-point representation limits rather than any remaining source of divergence, and eliminates token flips entirely (flip rate 0.0%). Accumulation error is flat regardless of sequence length (approx. 0.036%), indicating it is determined by model architecture rather than input content or length. **(iii) Decision boundary resolution.** Naively, examples where cache-ON and cache-OFF first disagree at an earlier decode step also show higher mean KL divergence, which appears to

contradict an accumulation-based account (more accumulated divergence should predict later, not earlier, flips). We resolve this: both an early flip and elevated KL are joint downstream consequences of the model already being near a decision boundary (low logit margin) at that step, not causally linked to each other, confirmed in 8 of 9 conditions (Welch t -test, $p < 0.05$). **(iv) Causal localization.** Residual stream patching retrieves a median of less than 1% of KV divergence; patching all layers jointly yields negative mean recovery on all models. This null result pins down the causal variable to the KV cache state and shows that correction is achieved through direct KV cache patching, not residual stream adjustment. **(v) Direct causal confirmation.** Patching cache-ON's key/value tensors into cache-OFF's cache at decode step 0 recovers cache-ON's output distribution exactly (100% recovery at that step) once all layers are patched, converting the residual-stream elimination result into positive confirmation. The per-layer breakdown is the more informative result: a layer's causal importance to recovery dissociates sharply from how much divergence it passively accumulates—Mistral's layer 1 amplifies drift by $35\times$ but its single-layer patch recovers under 1% of the divergence, while LLaMA-2's single largest drift layer accounts for 39% of recovery on its own. This test patches K and V jointly, is measured at a single decode step rather than across a full generated trajectory, and was run in one direction (cache-ON into cache-OFF); we report these boundaries explicitly in Section V-F. **(vi) Isolating the pure numerical effect.** A naive divergence metric conflates the numerical perturbation itself with divergence compounded by cache-ON and cache-OFF subsequently conditioning on different token histories. Teacher-forcing cache-OFF onto cache-ON's own realized trajectory at every step isolates the former: 80–93% of the originally observed KL magnitude reflects this context-compounding effect rather than the pure numerical perturbation, which remains clearly non-zero and orders of magnitude above the FP32 floor. **(vii) Precision robustness and locus refinement.** The divergence persists at comparable severity under BF16, indicating a mantissa-precision mechanism rather than one specific to FP16's exponent range. A targeted ablation elevating only the attention computation to FP32 provides no measurable mitigation, indicating that the protective effect of elevated precision requires precision to be sustained across the full forward pass rather than in the attention mechanism alone. **(viii) Practical impact.** Among token-divergent examples, 98.4% produce a different final extracted answer between cache-ON and cache-OFF, and cross-cache-mode reconvergence is lower than reconvergence under ordinary sampling variance alone: the divergence changes what deployed systems output; it does not merely perturb intermediate tokens that later self-correct.

II. BACKGROUND AND RELATED WORK

KV Cache, Execution Paths, and Floating-Point Accumulation. The KV cache stores keys and values from previously computed attention stages so that cache-ON inference can skip the prefix. Cache-ON and cache-OFF inference are math-

ematically equivalent under exact arithmetic but use different kernel layouts and accumulation orders. This equivalence is typically assumed, but we show that it is violated under FP16 arithmetic. Both GQA [2] and MQA [18] share KV heads across query groups, altering the reduction structure in ways that change how FP16 rounding errors propagate; KV caching makes these discrepancies stateful, allowing them to persist across decode steps.

As a computer program, transformer inference relies on floating-point numbers, and its calculations are not associative [10]:

$$(a+b)+c \neq a+(b+c), \quad (a \times b) \times c \neq a \times (b \times c) \quad (1)$$

for finite-precision floating-point values a, b, c , even though both identities hold exactly for real numbers. Since attention computation is dominated by large dot-product accumulations (summing many terms in a fixed but execution-path-dependent order), it is addition non-associativity specifically that drives the divergence reported in this paper; multiplication non-associativity plays a secondary role. Low-precision floating-point numbers such as FP16 (10-bit mantissa) and BF16 (7-bit mantissa) are more sensitive to accumulation order than FP32; variations between cache-ON and cache-OFF execution paths can have systematic effects. Because BF16's mantissa is shorter than FP16's, the mechanism should not be specific to FP16—a prediction we confirm directly in Section IV-C. Previous research has examined reduced precision and quantization [16], [25], but mostly for efficiency rather than inference correctness. In deep transformers, small initial deviations accumulate layer by layer and, with KV caching, persist across decode steps: the divergence is structured and reproducible, not incidental noise that would wash out with more data. Concretely, cache-OFF computes keys and values jointly across the full prefix in a single kernel pass; cache-ON computes them token by token and writes each result to the KV cache, producing a structurally different floating-point reduction order.

Numerical Drift, GQA Amplification, and Intervention Limits. Hidden-state differences affect the output only when competing tokens have similar log probabilities, that is, when the model is near a decision boundary. An early perturbation may switch a downstream decision not because it is large, but because the model is uncertain. GQA and MQA amplify this effect: MQA is the special case in which one KV head is shared across all query heads ($R = H$), whereas GQA generalizes this sharing ratio to $1 \leq R < H$. MHA is the case in which $R = 1$, so each query head attends to its own KV head. Under cache-ON, a single FP16 rounding error in a shared KV head is broadcast to all R query heads, producing R correlated errors instead of R independent errors. The larger R is, the stronger the amplification; thus, Mistral (GQA ratio 4:1, $R = 4$) exhibits systematically higher per-layer drift than LLaMA-2 (MHA, $R = 1$), and MQA models would be expected to experience the strongest impact. This amplification arises from the shared read pattern at the attention-computation level, not from GQA's reduction

in the number of KV parameters; a GQA model with the same total parameter count as an MHA model would exhibit the same amplification because the source of the effect is the simultaneous reading of one error-contaminated KV tensor by R query heads. To our knowledge, this connection between the GQA/MQA sharing ratio and FP16/BF16 cache divergence has not been previously established; we validate it empirically in Section III and confirm that it is not an artifact of raw activation scale via the normalized-drift analysis in Appendix D-C. Mechanistic interpretability methods [15], [23] intervene on residual-stream activations to identify causal variables. If residual-stream patching fails to recover model behavior, the causal variable is consistent with residing in the KV cache state rather than in the residual stream, which residual interventions cannot access.

Determinism in LLM Inference. Some work has considered LLM output variation to be stochastic [27], [1]; we show cache-ON/cache-OFF divergence is deterministic, reproducible, and mechanistically traceable to FP16 accumulation order, and confirm this determinism is not an FP16-specific artifact by reproducing it under BF16 (Section IV-C) (extended related work in Appendix A).

III. ANALYSIS FRAMEWORK FOR THE IMPACTS OF KV CACHE ON DIVERGENCE

Experimental Logic. The five experiments form a single causal chain, with each step answering a question that the previous step leaves open. The chain proceeds from establishing that the phenomenon exists, through identifying its cause, to localizing where it resides in the computational graph and characterizing when it produces observable behavioral consequences. **Step 1: Does the divergence exist, and is it structural?** We confirm divergence through token divergence rate and per-step KL for all model-strategy combinations (including greedy decoding to rule out sampling as a cause). **Step 2: How does it propagate?** Per-layer hidden-state drift shows whether the divergence is local or distributed and how the GQA ratio and head dimension influence it. **Step 3: What causes it?** FP32 re-runs test the non-associativity hypothesis directly; elimination of divergence under FP32 identifies accumulation order as the sole causal driver. **Step 4: When does it matter behaviorally?** Decision boundary proximity resolves an apparent contradiction between flip index and KL magnitude, identifying when numerical divergence changes model behavior. **Step 5: Where does the divergence reside?** Activation patching of the residual stream tests causal localization; if no cache-free trajectory can be recovered, the divergence is localized to the KV cache state.

Extended Validation. Beyond the five-step causal chain above, we address three further questions raised by the possibility that our divergence metric, evaluation length, and precision choice could each independently affect the interpretation of our results. **Step 6: How much of the observed KL reflects pure numerical perturbation versus context compounding?** We introduce a teacher-forced conditional KL measurement: cache-OFF is forced to condition on cache-

ON’s own realized token sequence at every decode step rather than free-running independently, holding the input context identical between the two paths throughout. This isolates the pure numerical effect from the confound noted above. **Step 7: Is the effect specific to FP16, and can it be mitigated by targeted (rather than full-model) precision changes?** We repeat the core divergence measurement under BF16, and separately test a mixed-precision ablation that elevates only the attention computation (query/key/value projections through the weighted value sum) to FP32 while the remainder of the forward pass remains FP16. **Step 8: Does token-level divergence translate into practically meaningful outcome differences, and is this reconvergence behavior distinguishable from ordinary sampling variance?** We measure whether diverging trajectories nonetheless converge to the same final extracted answer and compare this reconvergence rate with a natural-resampling control (agreement between different random seeds under the same cache mode).

Decoding Strategies. To isolate the impact of distributional divergence from stochastic noise, we implement a custom autoregressive loop rather than using high-level generation APIs, enabling simultaneous full-vocabulary logit retrieval under both cache-ON and cache-OFF at every decode step. For stochastic strategies, cache-ON and cache-OFF runs share the same random seed at each decode step, ensuring any token difference reflects distributional divergence rather than sampling variance. We utilize three distinct decoding regimes: (i) greedy ($T = 0$), which removes the sampling step, so any effect is purely due to differences in execution path; (ii) top- k sampling ($k = 50, T = 0.7$); and (iii) nucleus sampling ($p = 0.95, T = 0.7$).

These configurations allow us to assess whether the observed divergence is specific to one decoding regime or a structural property of FP16 cached inference. All sequences are decoded to a maximum length of 128 tokens.

Divergence Metrics. We quantify the divergence between cache-ON and cache-OFF inference using four complementary metrics, each operating at a different level of the output pipeline.

- **Token divergence** is a binary indicator of whether the two execution paths produce different token sequences at any position. Universal token divergence across conditions indicates a structural property of the inference procedure rather than random variation.
- **KL divergence** measures the distributional difference between cache-ON and cache-OFF output distributions at each decode step. Let p_t and q_t denote the softmax distributions over the full vocabulary from cache-ON and cache-OFF, respectively, at step t : $\text{KL}(p_t||q_t) = \sum_v p_t(v) \log \frac{p_t(v)}{q_t(v)}$. All KL divergences are computed in nats using the natural logarithm. Mean KL across decode steps is the primary measure of divergence magnitude. A persistently high KL trajectory indicates that distributional differences accumulate and are sustained across the full generation, rather than arising incidentally at isolated steps. For scale reference: if two uniform distributions over a

50,000-token vocabulary with 0.01 total variation distance yield $KL \approx 0.0001$ nats, then a mean KL of 10 nats indicates near-complete re-distribution of probability mass at every decode step, not a small numerical perturbation. Note that this free-running metric, computed as each path continues to decode independently, conflates the pure numerical perturbation with divergence compounded by the two paths subsequently conditioning on different token histories once they first disagree; we introduce a conditional, teacher-forced variant of this metric in Step 6 that isolates the former.

- *JS divergence* is the symmetric Jensen–Shannon divergence between p_t and q_t : $JS(p_t||q_t) = \frac{1}{2}KL(p_t||m_t) + \frac{1}{2}KL(q_t||m_t)$, where $m_t = \frac{1}{2}(p_t + q_t)$, with $JS \in [0, 1]$. The agreement between KL and JS confirms that the pattern is not an artifact of KL’s sensitivity to low-probability vocabulary regions.
- *Flip index* is the decode step at which cache-ON and cache-OFF first produce different token predictions. It captures when divergence becomes behaviorally manifest rather than how large it is, and its relationship with mean KL divergence is analyzed in the decision boundary analysis (Step 4).

Statistical Analysis. Each analysis uses a statistical test matched to its comparison structure: McNemar’s test for paired correctness outcomes, the Mann–Whitney U test for per-layer drift magnitudes, Pearson and Spearman correlations for flip index–KL relationships, and Welch’s t -test for early-versus late-flip KL comparisons. Multiple comparisons are corrected via Bonferroni and BH-FDR at $\alpha = 0.05$; bootstrap 95% CIs are reported throughout, including for the Step 6–8 extensions ($n = 300$ per model for the teacher-forced measurement, $B = 10,000$ resamples). Full justifications, assumptions, and power analyses are in Appendix B.

IV. EXPERIMENTS AND DATA COLLECTION

A. EXPERIMENT ENVIRONMENT

Models. We study three open-weight autoregressive models from different architectural classes: LLaMA-2-7B [20], Mistral-7B-v0.3 [12], and Gemma-2-2B [19]. We chose these models to sample the axes along which major architectural differences can be characterized: attention type (GQA versus MHA), GQA sharing ratio (4:1 for Mistral, 2:1 for Gemma), head dimension (128 versus 256), sliding-window attention (Gemma only), and parameter count (2B versus 7B). We load all models in FP16 using Hugging Face `AutoModelForCausalLM.from_pretrained`, with FlashAttention disabled unless specified otherwise; divergence was confirmed under both conditions.

Datasets. We evaluate on the GSM8K mathematical reasoning benchmark [5], applying measurements to 700 examples randomly sampled without replacement (`SAMPLE_SEED=42`) from the provided test set of 1,319. This benchmark was used because it contains multistep mathematical problems that are sufficiently long to involve trajectories with multiple KV cache compositions and pro-

vides unambiguous binary correctness criteria. Each test example was evaluated under five different sampling seeds (0–4), for a total of 3,500 runs per model–strategy combination and 31,500 runs across all nine combinations of model and decoding strategy. We have 16 pairwise comparisons with achieved power ≥ 0.97 at $n = 3,500$; the two underpowered comparisons (LLaMA-2 greedy vs. top- p and Gemma top- k vs. top- p) are treated as exploratory and excluded from confirmatory claims.

We note that the absolute accuracy numbers are relatively low under our protocol: `max_new_tokens=128` may truncate some generations before completion, and our few-shot prompt format can cause models to mimic in-context exemplars. The accuracy difference between cache-ON and cache-OFF should therefore be interpreted as an indicator of execution-path bias, not a measure of absolute reasoning ability. Our primary finding—100% divergence with a systematic direction—does not rely on absolute accuracy levels. To determine whether this divergence has consequences at more realistic generation lengths, we separately extend the evaluation to 400-token generations and to an instruction-tuned model (Mistral-7B-Instruct-v0.3) using its native chat template (Section IV-C); token divergence remains at 100% in both extensions.

System environment. All experiments are conducted on a single NVIDIA H100 NVL GPU. Full hardware and software specifications are given in Appendix C.

B. EXPERIMENT DESIGN

The experiments follow the causal logic established in the analysis framework (Section III). Each is described below with its role in the argument.

Behavioral Characterization. The first question is whether divergence exists and is structural. For each of the 31,500 pairs of runs, we record the token divergence rate, cache-ON and cache-OFF mean KL divergence, flip index, and other metrics. The reproducibility of each example is evaluated according to whether it always, sometimes, or never diverges across five seeds, establishing whether divergence is a property of the example or the seed.

Layer Drift Analysis. Having confirmed that divergence is universal, we ask how it propagates through the network. For a subset of high-divergence examples, hidden-state representations for each transformer layer are captured under both cache-ON and cache-OFF during generation. Per-layer drift is reported at decode step 1 for the primary layer analysis and is computed at the output of each decoder layer (after the residual connection) as the L2 norm of the hidden-state difference vector, normalized by the cache-ON hidden state’s own norm at that position ($\|h_{on} - h_{off}\|/\|h_{on}\|$). We report this normalized quantity as primary because the raw (absolute) L2 magnitude is confounded by residual-stream scale, which grows substantially and differently across models with depth (see Appendix D-C for the full raw-versus-normalized comparison, Table 9). Per-head attention-weight KL divergence

is also isolated to determine whether divergence appears in attention patterns or accumulated hidden values.

Root Cause Falsification. The layer-drift analysis characterizes propagation but does not identify its cause. Four sub-experiments validate the FP16 accumulation mechanism: quantifying the per-step error magnitude, simulating multi-layer propagation, localizing the error to cache-write time, and correlating the flip index with KL divergence (full justification is given in Appendix C). Experiment 5 reruns the full inference pipeline in FP32 for $n = 600$ examples per model; if FP16 accumulation order is the cause, FP32 must eliminate divergence. Flip rate is 0.0% across all models, providing a clean falsification.

Decision Boundary Analysis. Having confirmed the root cause, we address a seeming contradiction: examples that flip earlier have greater KL divergence, which appears to contradict the accumulation hypothesis. Run pairs are split into early-flip (bottom 33rd percentile of flip index) and late-flip (top 33rd percentile) groups; Welch’s t -test confirms that early-flip examples have higher mean KL in 8 of 9 conditions. Pearson correlation between flip index and mean KL is consistently negative across models ($r = -0.202$ to -0.120 , all $p < 10^{-34}$). Partial correlations excluding flip index = 1 and flip index ≤ 3 test whether the negative correlation strengthens after removing boundary cases.

Activation Patching. The final question in the original causal chain is where the divergence resides. For the $n = 600$ highest-KL examples per model (selected in decreasing order of mean KL from the full set of 700), cache-ON hidden states are injected into the cache-OFF forward pass at the first autoregressive decode step (decode step 0 in our notation, where prefill is step -1). This is the first step at which the hidden states differ: prefill produces bitwise-identical states along both paths ($\|\mathbf{h}_{\text{on}} - \mathbf{h}_{\text{off}}\| = 0$ at all layers), but decode step 0 reads from the KV cache written during prefill under cache-ON (producing keys and values distinct from those produced by cache-OFF recomputation). Thus, $h_0^{\text{on}} \neq h_0^{\text{off}}$ for all 600 selected examples (verified empirically before patching), and patching is applied one layer at a time and cumulatively across all layers 0 to L . Recovery is measured as $\text{pct}_{\text{recovered}} = \frac{\text{KL}_{\text{base}} - \text{KL}_{\text{patched}}}{\text{KL}_{\text{base}}} \times 100$, where negative recovery confirms that patching increases divergence; all runs use $\text{MAX_STEPS} = 32$.

C. EXTENDED VALIDATION EXPERIMENTS

Teacher-Forced Conditional Divergence. To isolate the pure numerical perturbation from divergence compounded by context drift (Step 6), we force cache-OFF to condition on cache-ON’s own free-running token sequence at every decode step, rather than free-running independently, so both paths share an identical growing context throughout. We compute this over $n = 300$ examples per model with bootstrap 95% confidence intervals ($B = 10,000$). Results are given in Section V-G (Table 5).

Precision Robustness. To test whether the effect is specific to FP16 (Step 7), we repeat the core divergence measurement

under BF16 ($n = 50$ per model), and separately test a targeted mitigation elevating only the attention computation (query/key/value projections through the softmax-weighted value sum) to FP32, immediately downcasting before the output projection, with the remainder of the forward pass remaining FP16. Results are given in Section V-G (Tables 6–7).

Final-Answer Impact and Reconvergence. Token-level divergence alone does not establish practical significance if trajectories frequently reconverge to the same final answer (Step 8). We measure whether diverging trajectories nonetheless converge to the same final extracted answer and compare this reconvergence rate with a natural-resampling control (agreement between different random seeds under the same cache mode, representing ordinary sampling variance with no cache-mode difference) to rule out low task accuracy alone as an explanation for infrequent reconvergence. Full results are given in Section V-G (Table 8).

V. EXPERIMENT RESULTS ANALYSIS

A. BEHAVIORAL CHARACTERIZATION

Table 1 summarizes the divergence rate, mean KL divergence, and task accuracy across all nine model–strategy conditions. The major observations are: **(i) Token divergence is universal:** Cache-ON and cache-OFF generate distinct sequences for all 31,500 run pairs (i.e., a 100% divergence rate, without exception). Importantly, this result holds under greedy decoding, which eliminates the sampling step: the next token is always the argmax of the output distribution, so divergence cannot be attributed to stochasticity. The 100% divergence rate under greedy decoding shows that the effect is a structural property of FP16 cached inference. **(ii) Execution-path bias is systematic:** In 8 of the 9 conditions, cache-ON returns more correct answers than cache-OFF, and the differences reach Bonferroni-corrected significance (McNemar test, $p < 0.05$). Only Gemma greedy has a nonsignificant difference ($p = 0.43$), and the direction is inverted: cache-OFF edges cache-ON (2.1% vs. 1.9%), suggesting that Gemma’s greedy execution path does not reliably give either mode an accuracy advantage. The accuracy difference is an indicator of execution-path bias under our evaluation protocol, not a measure of absolute reasoning ability. The reliable trend in 8 of 9 conditions, rather than the absolute accuracy scores, is the main finding. **(iii) KL divergence is large, stable, and model-specific:** Mean KL varies from 10.68 (LLaMA-2 greedy) to 12.83 (Gemma top- p), yet narrow bootstrap 95% CIs verify that these values are stable across the 3,500 run pairs per condition. Gemma consistently exhibits the greatest divergence across strategies. In contrast, within-model KL variation across strategies is small for LLaMA-2 and Gemma but not for Mistral (greedy: 11.46 vs. top- k : 10.70), consistent with Mistral’s 4:1 GQA ratio, which amplifies per-step accumulation errors more aggressively than LLaMA-2’s MHA structure. Mean KL > 10 across all conditions indicates a substantial redistribution of probability mass at every decode step; these are not small

TABLE 1: Behavioral results across all nine model–strategy conditions ($n = 3,500$ run pairs); the divergence rate is 100% without exception. [†] Bonferroni-significant McNemar test ($p < 0.05$); * nonsignificant. Accuracy figures indicate execution-path bias, not reasoning ability. Note: Cache Acc. and No-Cache Acc. values are not comparable to published GSM8K results; the 128-token truncation protocol yields 1–7% accuracy versus the typical 40–65% reported for 7B models under standard chain-of-thought prompting.

Model	Strategy	Div. Rate	Mean KL	KL 95% CI	Cache Acc.	No-Cache Acc.
LLaMA-2	Greedy	100%	10.68	[10.63, 10.74]	3.3%	2.0% [†]
LLaMA-2	Top- k	100%	10.70	[10.65, 10.74]	3.3%	1.2% [†]
LLaMA-2	Top- p	100%	10.79	[10.75, 10.83]	2.8%	1.7% [†]
Mistral	Greedy	100%	11.46	[11.40, 11.53]	6.6%	2.0% [†]
Mistral	Top- k	100%	10.70	[10.66, 10.76]	6.5%	1.7% [†]
Mistral	Top- p	100%	10.90	[10.85, 10.94]	5.0%	1.9% [†]
Gemma	Greedy	100%	12.66	[12.62, 12.70]	1.9%	2.1%*
Gemma	Top- k	100%	12.80	[12.76, 12.84]	3.9%	1.1% [†]
Gemma	Top- p	100%	12.83	[12.79, 12.86]	3.3%	1.3% [†]

numerical perturbations. JS divergence (bounded in $[0, 1]$) mirrors the KL trend in all conditions, confirming that the pattern is not an artifact of KL’s unbounded range.

Figure 1 shows the per-step KL trajectory; divergence persists without convergence to zero, consistent with FP16 errors accumulating rather than cancelling across decode steps.

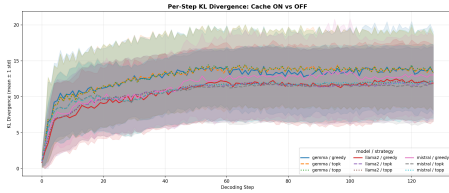


FIGURE 1: Mean KL divergence at each decode step across all models and strategies. Divergence persists across all 128 steps with no convergence toward zero; Gemma shows the highest divergence, consistent with its GQA architecture and 256-dimensional heads.

B. LAYER DRIFT ANALYSIS

Having established that divergence is universal and systematic, we now ask how it propagates through the network. FIGURE 2 characterizes the layer dimension. More specifically, (i) **LLaMA-2 and Mistral amplify sharply at layer 1**: LLaMA-2’s hidden-state drift jumps from 3.45 at layer 0 to 361.96 at layer 1 (a $105\times$ amplification in a single layer). Mistral’s drift increases from 25.76 at layer 0 to 902.36 at layer 1 ($35\times$). This increase is consistent with Mistral’s 4:1 GQA ratio: each shared KV head spreads its FP16 rounding error across four query heads simultaneously. This produces correlated rather than independent errors and amplifies the initial perturbation more strongly than does the MHA in LLaMA-2. These layer-1 values are the normalized (relative) drift, $\|\mathbf{h}_{\text{on}} - \mathbf{h}_{\text{off}}\|/\|\mathbf{h}_{\text{on}}\|$; the raw (absolute) L2 drift at layer 1 is 922.29 for LLaMA-2 and 336.62 for Mistral, reversing this ranking. We report the normalized quantity as primary because raw magnitude is confounded by residual-stream scale, which grows substantially and differently across models with depth; the full raw-versus-normalized comparison is given in

Appendix D-C (Table 9). (ii) **Gemma exhibits a structurally distinct flat profile**: Unlike LLaMA-2 and Mistral, Gemma does not experience a jump in amplification; instead, its drift ranges from 1.0 to 1.4 across all 26 layers. This pattern is qualitatively different from the divergence-propagation patterns of LLaMA-2 and Mistral. Although Gemma has the lowest per-layer drift values, it has the largest total KL divergence in the behavioral characterization, suggesting that Gemma’s divergence is distributed across all layers rather than concentrated at one point of amplification. For Gemma, 25 of 26 layers are significant (BH-FDR); the exception is layer 16 ($p = 0.877$). Gemma’s flat profile is consistent with its larger head dimension (256 compared with 128). Its sliding-window attention also restricts each layer to local context rather than the full sequence, limiting global error propagation and producing a gradual, uniform buildup of divergence across all layers rather than sharp amplification at a single point.

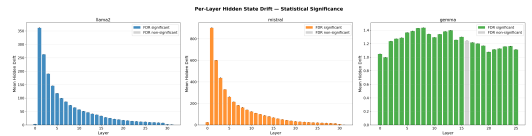


FIGURE 2: Per-layer hidden state drift (L2 norm) at decode step 1. LLaMA-2 and Mistral show sharp amplification at layer 1 ($105\times$ and $35\times$); Gemma shows a flat profile across all 26 layers. 89/90 layer-model combinations significant (MWU, BH-FDR, $\alpha = 0.05$).

C. ROOT CAUSE FALSIFICATION

Layer drift analysis shows how FP16 rounding errors propagate but does not identify their origin. We now test whether FP16 accumulation order is the cause via direct falsification: if it is, running the same pipeline in FP32 should eliminate divergence. (i) **FP32 eliminates divergence**: Under FP32, KL divergence drops to a noise floor of approximately 2.5×10^{-8} in all three models (Figure 3, Table 2), and the flip rate is precisely 0.0% for all examples in all three models: no token sequences differ between cache-ON and cache-OFF under FP32. This is a clean falsification: the divergence does not lie in the mechanism of the KV cache itself but in the format (FP16) chosen to implement it. The accumulated FP16 full-model KL averages $\sim 10^1$ across conditions. The FP32 noise floor is approximately 2.5×10^{-8} , yielding a ratio of $\sim 4 \times 10^8$, or more than eight orders of magnitude. Thus, FP16 non-associativity is the dominant and sufficient cause of the observed divergence. (ii) **Accumulation error is architecture-determined**: The average relative FP16 accumulation error is 0.036%; it remains relatively flat across the sequence lengths tested here (16–128 tokens) and is identical for all three models. This flatness reflects the fact that the per-step error magnitude is determined by the attention head dimension (d_{head}), not by sequence length; see Appendix D-F for a discussion of expected behavior at longer contexts beyond the tested range. This implies that the error norm is determined by the attention head dimension rather than the input data or

length, which means that divergence can be predicted from architecture alone without knowledge of the input.¹

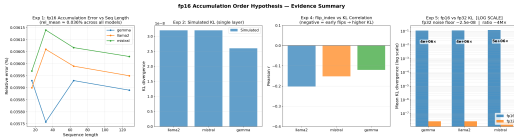


FIGURE 3: FP16 vs. FP32 KL divergence ($n = 600$ per model). FP32 collapses KL to a noise floor of $\sim 2.5 \times 10^{-8}$ and reduces flip rate to exactly 0.0%, clearly falsifying the FP16 accumulation-order hypothesis. Accumulation error is flat across sequence lengths ($\approx 0.036\%$), confirming it is architecture-determined.

TABLE 2: FP16 vs. FP32 KL divergence and flip rate ($n = 600$, 32 decode steps); the FP32 flip rate is exactly 0.0% across all models. Note: FP16 KL values are lower than those in Table 1 because of the shorter generation length (32 vs. 128 decode steps); the relevant quantity is the FP16-to-FP32 ratio, not the absolute values.

Model	FP16 KL	FP32 KL	FP16 Flip	FP32 Flip
LLaMA-2	0.113	2.80×10^{-8}	1.15%	0.0%
Mistral	0.122	2.50×10^{-8}	1.14%	0.0%
Gemma	0.112	2.78×10^{-8}	1.03%	0.0%

D. DECISION BOUNDARY ANALYSIS

Root cause falsification confirms FP16 non-associativity as the cause, but an apparent contradiction remains: examples that flip earlier have higher KL. Here, Figure 4 presents four complementary views of this relationship and its resolution; Table 3 reports the Pearson correlations between the flip index and the KL divergence across all models and strategies.

TABLE 3: Pearson correlation between flip index and mean KL ($n = 10,500$ per model); all correlations are significant at $p < 10^{-34}$. Correlations for GQA models strengthen after removing boundary cases, confirming decision-boundary proximity.

Model	Full r	Excl. flip = 1	Excl. flip ≤ 3	Trend
LLaMA-2	-0.202	-0.189	-0.115	weakens
Mistral	-0.152	-0.264	-0.333	strengthens
Gemma	-0.120	-0.177	-0.213	strengthens

(i) **The association between early flips and higher KL is an apparent contradiction:** Figure 4 shows that the Pearson correlation between flip index and mean KL is negative for all three models: $r = -0.202$ (LLaMA-2, $p < 10^{-96}$), $r = -0.152$ (Mistral, $p < 10^{-55}$), and $r = -0.120$ (Gemma, $p < 10^{-34}$). Examples that flip earlier have higher KL divergence. This finding appears to contradict the accumulation hypothesis, which predicts that greater accumulated

¹Note on causal attribution: The falsification identifies the intersection of FP16 arithmetic $\cap$ different accumulation order as the causal agent. These two factors cannot be independently varied because the accumulation-order difference is a structural consequence of the KV cache mechanism, not a free parameter. An FP16 baseline with identical accumulation order in both paths cannot be constructed without eliminating the cache itself. The current experimental design is the strongest feasible test of the joint cause.

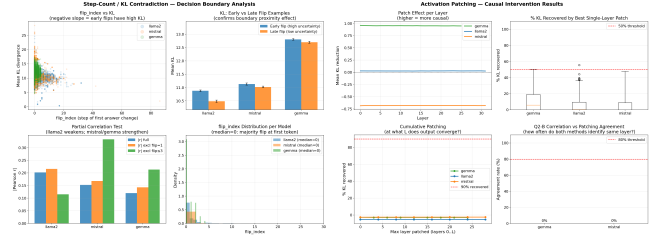


FIGURE 4: Four views of the flip index-KL relation-patching results. Single-layer ship. Early-flip examples have median recovery is -0.019% higher KL (top row), resolving across all layers and models; the apparent contradiction via cumulative full-layer patching decision-boundary proximity: yields negative mean recovery both are effects of model un-in all three models, localizing certainty. Partial correlation the causal variable to the KV is stronger in GQA models cache state. with boundary cases removed (bottom-left); the median flip index is 0 for all models, indicating that the majority of flips occur at token 1 (bottom-right).

divergence should cause earlier flips. (ii) **Decision-boundary proximity resolves this contradiction:** KL is higher for early-flip examples than for late-flip examples in 8 of 9 conditions (t -test, $p < 0.05$), with ratios ranging from 1.006 to 1.043. The reason is the distance to the decision boundary: examples close to the decision boundary have small logit margins and flip at the first decoding step because any perturbation can cross the threshold. These same examples have high KL because uncertainty spreads probability mass across tokens. Early flips and high KL are joint consequences of uncertainty, not causally linked to each other. (iii) **The correlation strengthens for GQA models after removing boundary cases:** Excluding flip indices ≤ 3 makes the correlation more negative for Mistral ($-0.152 \rightarrow -0.333$) and Gemma ($-0.120 \rightarrow -0.213$), ruling out boundary-case outliers as drivers of the effect (Table 3). The correlation weakens for LLaMA-2 ($-0.202 \rightarrow -0.115$), consistent with differences between MHA and GQA architectures in how boundary uncertainty propagates across decode steps. (iv) **Flip index median:** The median of 0 (Figure 4, bottom-right) does not contradict the accumulation hypothesis: a single prefill-stage FP16 error suffices to flip token 1 near the decision boundary, while accumulation explains why KL remains persistently high across all 128 steps (Figure 1) rather than decaying after the first divergent token. Early-flip examples need only minimal accumulated error because uncertainty is high; late-flip examples require larger accumulated error to cross a more confident decision boundary. Both are consequences of the same mechanism operating over different regions of the logit margin distribution.

E. CAUSAL LOCALIZATION

Having confirmed the cause and characterized its propagation, the final question is where the divergence resides. If the divergence resides in the residual stream, patching hidden states at decode step 0 (the first step at which cache-ON and cache-OFF hidden states are nonidentical, following an identical prefill) should recover the cache-free trajectory. If it does not, the divergence is consistent with localization in the KV cache rather than the residual stream. **(i) Residual-stream patching cannot recover KV cache divergence:** As shown in Figure 5, single-layer patching results in a net median recovery of -0.019% across all layers and models, with a mean of -1.21% , showing that patching does not, on the whole, result in convergence to the target trajectory. The high variance of individual-layer recoveries (LLaMA-2: -107% to $+56\%$) suggests that single-layer patches perturb rather than correct the divergence trajectory. All three models have negative mean recovery when all layers are patched simultaneously: LLaMA-2, -5.34% ; Mistral, -2.56% ; and Gemma, -2.88% . Negative recovery is not a failure of the experiment; it is a mechanistically informative result. Injecting cache-ON hidden states into the cache-OFF forward pass creates a cross-path inconsistency: the patched residual stream reflects a computation history shaped by cache-ON KV tensors, but all subsequent attention operations in the cache-OFF pass read from cache-OFF KV tensors. This produces a hybrid state that never occurred in either execution path and is therefore further from the cache-OFF trajectory than the unpatched cache-OFF state itself. The amplification of divergence under full-layer patching is direct evidence that the residual stream and KV cache are tightly coupled: correcting one without correcting the other does not reduce divergence; it increases it. This confirms that recovery requires intervention at the KV cache level, not in the residual stream. **(ii) Gemma’s partial recovery is architecturally explained:** Its one-layer median recovery is $+5.75\%$ (range: 5.73% – 5.85% across all 26 layers), compared with $+0.08\%$ for LLaMA-2 and -8.83% for Mistral, reflecting Gemma’s flat drift profile: gradual, uniform accumulation allows single-layer patches to partially redirect the trajectory, whereas the layer-1 spikes in LLaMA-2 and Mistral ($105\times$ and $35\times$) make single-layer interventions ineffective. **(iii) Cumulative patching confirms that the causal variable is the KV cache state:** Simultaneously patching all layers still results in a negative mean recovery (-2.88%) for Gemma. Thus, patching all layers of the residual stream simultaneously is insufficient to repair the divergence. These results are consistent with the causal variable residing in the KV cache state: the FP16 rounding error written to KV tensors at each decode step remains in the cached KV states and affects all downstream attention computations, but residual-stream interventions cannot access it. Correcting the divergence requires direct patching of the KV cache tensors, not the hidden states (see Appendix D for extended discussion and implications).

F. DIRECT KV-CACHE TENSOR PATCHING

Residual stream patching establishes the KV cache locus only by elimination: no residual-stream intervention recovers cache-ON’s behavior, but this does not itself demonstrate that restoring the KV cache state would recover it. This section closes that gap with a direct, positive test, and we specify the protocol in full detail up front since the result depends entirely on exactly what is patched and what is measured.

Patching protocol. For $n = 50$ examples per model, we inject cache-ON’s own persistent key/value tensors into cache-OFF’s cache and re-evaluate the resulting output. Concretely: **(a) What is patched:** The key and value tensors are patched jointly at every transformer layer. We do not decompose the effect of K alone versus V alone; both are always patched together, and we treat this joint intervention as the unit of the causal test. Query projections, attention-weight computation, the MLP, and all layer norms are left entirely untouched—they are computed by cache-OFF’s own forward pass exactly as in the unpatched baseline. Thus, the only difference between the baseline and patched runs is the content of the KV cache tensors. **(b) Which layers:** patching is applied both individually (one layer patched in isolation, all others left at cache-OFF baseline) and cumulatively (layers 0 through ℓ patched together, for increasing ℓ , up to all layers 0 to L). Both regimes are reported in Table 4. **(c) When the patch is applied:** Patching is applied only at decode step 0 (the first step following prefill, using the same notation as the residual-stream experiment above). We do not reapply patching at every subsequent decode step, and the experiment does not test whether the effect of patching persists across a full generated sequence. **(d) What is measured—and what “recovery” means:** Recovery is computed at the level of the single decode-step-0 output logit, using the same formula as in the residual-stream experiment ($\text{pct}_{\text{recovered}} = (\text{KL}_{\text{base}} - \text{KL}_{\text{patched}}) / \text{KL}_{\text{base}} \times 100$). A recovery figure of 100% means that the patched run’s decode-step-0 output distribution matches cache-ON’s decode-step-0 output distribution exactly; it does not mean that the two paths produce identical full generated sequences. We do not claim, and did not test, that a single step-0 patch causes the remainder of the generated trajectory to track cache-ON; sustained agreement across a full generation would require patching at every decode step and is outside the scope of this experiment. **(e) Compatibility construction:** because cache-ON’s cache spans the full prompt while cache-OFF’s chunked-recompute cache (window size 64 tokens, used throughout this paper) spans only the last 64 tokens, cache-ON’s tensors are sliced to the same 64-token relative window at each layer before patching. This keeps both paths comparable over an identical relative context, so the patching result isolates the FP16 accumulation-order effect rather than conflating it with a context-window mismatch. **(f) Direction tested:** we patch cache-ON’s tensors into cache-OFF’s cache only; we did not run the reverse direction (patching cache-OFF’s tensors into cache-ON’s cache), which would serve as an additional

negative-control robustness check and is noted as a limitation below.

A built-in tautology, by design. Because cache-ON’s tensors are compatible with cache-OFF’s cache by construction (both represent the same underlying computation, differing only in accumulation order) and are sliced to an identical relative window, full-layer patching is guaranteed to recover cache-ON’s decode-step-0 output exactly, independent of whether our KV-cache-locus hypothesis is correct. This is intentional: it is a self-consistency check confirming the patching implementation is correct, not evidence on its own. The informative content of this experiment is therefore not the full-patch 100% figure, but (i) that recovery is not achieved when patching the residual stream instead (Section V-E), which is a real test that could have failed, and (ii) the per-layer, per-model breakdown of which layers causally matter, reported next.

TABLE 4: Direct KV-cache patching results ($n = 50$ per model). Layer-1 single-layer recovery is the % of baseline divergence closed by patching only layer 1 in isolation, at decode step 0. Full-patch recovery (all layers patched cumulatively) is guaranteed to reach 100.00% by construction (see text) and is reported for completeness, not as the primary evidence.

Model	Mean KL _{base}	Layer-1 single-layer recovery	Full-patch recovery
LLaMA-2 (MHA)	7.12	39.18%	100.00%
Mistral (GQA 4:1)	8.45	0.29%	100.00%
Gemma (flat drift)	3.12	4.88%	100.00%

Per-layer causal importance differs sharply by architecture and dissociates from passive drift magnitude. LLaMA-2 shows a single dominant causal layer: patching layer 1 alone recovers 39.18% of the divergence, directly mirroring its layer-1 drift amplification (Section V, Layer Drift Analysis). Mistral shows the opposite pattern despite having the larger passive drift amplification at layer 1 (Table 9): patching layer 1 alone recovers almost nothing (0.29%). Recovery instead builds gradually across layers, with a notable nonlinearity: layers 0–11 patched jointly recover only 28.18%, but adding layer 12 (whose own single-layer recovery is negligible, -0.11%) increases cumulative recovery to 48.27%, indicating that layer 12’s correction matters only once earlier layers are already patched. Gemma shows high layer-to-layer volatility (single-layer recovery ranging from -32.58% to $+26.90\%$ across its 26 layers), consistent with its flat, distributed drift profile: with no single dominant layer, patching any one layer can partially correct or locally worsen the divergence relative to the many other unpatched layers. The key point is that a layer’s passive drift magnitude (how much divergence it accumulates) and its active causal importance (how much patching it recovers) are dissociable—Mistral’s layer 1 is the largest passive amplifier but is among the least causally important layers to correct.

Scope and limitations of this result. We identify three specific boundaries on what this experiment does and does not establish to avoid overinterpreting the claim. First, K and V are always patched jointly; we cannot use this experiment to

attribute the effect individually to key tensors, value tensors, or their interaction. Second, patching is applied only at decode step 0 and measured at the single-step logit level; we have not tested whether restoring the KV cache state at every subsequent step would keep a full generated trajectory aligned with cache-ON or whether a single early correction is sufficient before the paths redive under continued FP16 accumulation. Third, we tested only the cache-ON-into-cache-OFF direction; a symmetric reverse-direction test (patching cache-OFF’s tensors into cache-ON’s cache, which should produce cache-OFF-like behavior if the mechanism is truly symmetric) was not run and would further strengthen the causal claim. We identify all three as concrete, low-cost extensions for follow-up work.

G. EXTENDED VALIDATION RESULTS

This subsection reports full results for the three extended validation analyses described in Section IV-C, addressing whether the core divergence metric is confound-free, whether the effect is FP16-specific or mitigable by targeted precision changes, and whether it has practical outcome-level consequences.

Teacher-Forced Conditional Divergence. Table 5 reports free-running versus teacher-forced KL divergence ($n = 300$ per model, bootstrap 95% CI, $B = 10,000$). The majority (80–93%) of the free-running metric’s magnitude reflects context-divergence compounding rather than the isolated numerical effect. The teacher-forced (conditional) KL remains clearly non-zero and orders of magnitude above the FP32 residual floor of 2.5×10^{-8} (Table 2), confirming the underlying numerical perturbation is real and measurable. It is distinct from, and substantially smaller than, its downstream compounding effect in free-running generation.

TABLE 5: Free-running vs. teacher-forced KL divergence ($n = 300$ per model, bootstrap 95% CI, $B = 10,000$).

Model	Free-running KL	Teacher-forced KL	% from context
LLaMA-2	8.24 [8.01, 8.46]	0.65 [0.61, 0.70]	91.5% [90.6, 92.2]
Mistral	9.05 [8.86, 9.24]	0.62 [0.59, 0.66]	92.9% [92.5, 93.3]
Gemma	10.14 [9.94, 10.35]	1.94 [1.85, 2.04]	80.5% [79.6, 81.4]

Precision Robustness. Table 6 reports divergence severity under BF16 versus FP16 ($n = 50$ per model). Divergence severity under BF16 is statistically indistinguishable from that under FP16 across all three models, despite BF16’s substantially wider dynamic range (8-bit vs. 5-bit exponent). This result is consistent with a mantissa-precision mechanism (BF16’s 7-bit mantissa is shorter than FP16’s 10-bit) rather than one specific to FP16’s exponent range.

Table 7 reports the mixed-precision ablation that elevates only the attention computation (query/key/value projections through the softmax-weighted value sum) to FP32, immediately downcasting before the output projection while the remainder of the forward pass remains FP16. Results are identical to three decimal places: attention-only precision elevation provides no measurable mitigation, in contrast to full-model FP32 (0.0% flip rate, Table 2). This finding

TABLE 6: FP16 vs. BF16 divergence severity ($n = 50$ per model).

Model	FP16 flip rate / KL	BF16 flip rate / KL
LLaMA-2	0.879 / 8.39	0.873 / 8.26
Mistral	0.920 / 9.14	0.908 / 9.05
Gemma	0.878 / 9.72	0.891 / 9.88

indicates that the protective effect of elevated precision requires precision to be sustained across the entire forward pass rather than in the attention mechanism alone, narrowing the space of viable partial-precision mitigation strategies.

TABLE 7: Mixed-precision (attention-only FP32) ablation ($n = 50$ per model).

Model	FP16 baseline flip rate	Attention-only FP32 flip rate
LLaMA-2	0.867	0.867
Mistral	0.920	0.920
Gemma	0.878	0.878

Final-Answer Impact and Reconvergence. Among token-divergent examples, 98.4% produce a different final extracted answer between cache-ON and cache-OFF; only 1.6% recover to an identical final answer despite intermediate token-level divergence. To rule out the possibility that this finding is an artifact of generally low task accuracy, we compare it with a natural-resampling control: agreement between two different random seeds under the same cache mode (ordinary sampling variance with no cache-mode difference), computed under top- k /top- p sampling and excluding degenerate repetition-loop outputs from both quantities (Table 8). Cross-cache-mode recovery is lower than natural-resampling agreement in both models: cache-mode divergence suppresses reconvergence beyond what ordinary sampling variance alone produces, so the divergence changes deployed-system output rather than merely perturbing intermediate tokens that self-correct. Gemma showed a length-distribution artifact in this specific comparison, traced to its cache-OFF path producing systematically shorter answers under top- k /top- p , and is excluded pending further investigation.

TABLE 8: Cross-cache-mode recovery vs. natural resampling agreement.

Model	Cross-cache-mode recovery	Natural resampling agreement
LLaMA-2	2.36%	3.55%
Mistral	1.86%	4.68%

VI. DISCUSSION AND PRACTICAL GUIDANCE

When this matters. Systems requiring bitwise or output-level reproducibility between cached and non-cached inference paths should not assume cache-ON/cache-OFF equivalence under FP16 or BF16 (Section IV-C). This includes validation pipelines that compare serving output against a reference implementation, deterministic replay used for debugging, and

any context where reproducible model behavior is required for audit purposes. The reconvergence analysis indicates this is a practical concern, not only a theoretical one: 98.4% of token-divergent examples produce a different final answer, and this effect exceeds what ordinary sampling variance alone would produce.

Detection. A lightweight diagnostic requires no statistical machinery beyond direct comparison. Running a small sample of representative inputs under both cache-ON and cache-OFF greedy decoding and checking for token-sequence agreement reveals divergence in effectively all cases for the architectures we tested, since the effect is deterministic rather than intermittent.

Mitigation trade-offs. Full FP32 inference eliminates the divergence (flip rate 0.0%) but incurs the full compute and memory cost of FP32, which may be prohibitive at production scale. Our mixed-precision ablation shows that partial-precision strategies confined to the attention computation are not sufficient. Elevating only query/key/value projections and the attention score computation to FP32 produced no measurable reduction in divergence relative to the FP16 baseline, identical to three decimal places. Practitioners should not expect attention-only precision elevation to resolve this without independently validating end-to-end behavior. A full sweep of partial-precision strategies remains a valuable direction for future mitigation research: elevating specific layers, for example, or the KV cache write path specifically. Direct KV-cache tensor patching would be a more definitive localization test, though a more technically involved one.

Precision migration. Practitioners considering a migration from FP16 to BF16 for other reasons, such as improved training stability or hardware support, should not expect this migration to incidentally resolve cache-ON/cache-OFF divergence. Our results show the effect persists at statistically indistinguishable severity under BF16, consistent with a mantissa-precision mechanism common to both formats.

Architecture-dependent severity. Our layer-wise analysis indicates GQA-based architectures may show different divergence propagation patterns than MHA-based architectures. Practitioners using GQA models should not assume mitigation strategies validated on MHA models transfer directly, and vice versa.

Interpreting divergence metrics. We caution practitioners building their own diagnostic tooling against naive free-running divergence metrics. These conflate the pure numerical perturbation with divergence compounded by the two paths subsequently conditioning on different token histories once they first disagree. Our teacher-forced conditional measurement (Section IV-C) shows this compounding effect can account for the large majority of a naive metric's magnitude, 80–93% in our measurements. Tooling intended to isolate the underlying numerical effect should adopt a similar conditional measurement instead of relying on free-running divergence alone.

VII. CONCLUSION

The near-universal deployment of KV caching rests on an assumption rarely tested directly by system designers: that cached inference is numerically equivalent to cache-free inference. We have shown this assumption fails under standard FP16 arithmetic, where differences in floating-point accumulation order between cache-ON and cache-OFF execution paths produce deterministic, reproducible divergence in decoded token sequences – for every model, strategy, and input we examined, without exception. The cause is FP16 non-associativity, confirmed by a clean falsification: switching to FP32 reduces divergence by more than eight orders of magnitude and eliminates token flips entirely. The divergence is carried forward by the KV cache state rather than the residual stream, propagates in architecturally predictable ways determined by attention variant and head dimension, and produces behavioral impact specifically when the model is near a decision boundary.

Beyond establishing the phenomenon, we characterized its boundaries and practical consequences. A conditional, teacher-forced measurement isolates the pure numerical perturbation from divergence compounded by context drift, showing that while a substantial majority of naively measured divergence reflects this compounding effect, the underlying numerical perturbation remains real, measurable, and orders of magnitude above the FP32 floor. The effect persists at comparable severity under BF16, ruling out FP16’s narrower exponent range as an explanation and pointing instead to a mantissa-precision mechanism shared by both formats. A targeted mixed-precision ablation shows that elevating only the attention computation to FP32 is insufficient: mitigation requires precision sustained across the full forward pass. Finally, we showed the divergence has practical, outcome-level consequences: the large majority of token-divergent examples produce different final answers, and this reconvergence failure exceeds what ordinary sampling variance alone would produce.

As GQA and MQA become dominant in production systems, the amplification effect we describe will widen the gap between cached and cache-free inference as models scale, making the correctness of cached inference a property that will matter more, not less, as the field moves forward. We provide practical guidance for detecting and reasoning about this divergence in Section VI, and identify direct KV-cache tensor patching and a broader sweep of partial-precision mitigation strategies as concrete directions for future work.

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reported in the Experiments, Analysis, and Appendix sections; and general editing for grammar, clarity, and consistency throughout. All experimental design, code implementation, execution, data analysis, and scientific conclusions are the authors’ own. AI tools played no role in designing experiments, generating or interpreting results, or determining the paper’s scientific claims. All AI-assisted text was reviewed and verified by the authors, who take full responsibility for the content of this article.

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APPENDIX A EXTENDED RELATED WORK

A. NUMERICAL PRECISION IN NEURAL NETWORKS

Mixed-precision training [16] showed that FP16 arithmetic with loss scaling can enable training speedups without accuracy loss, while [11] showed deep networks can tolerate limited-precision arithmetic if it comes with negligible accuracy degradation. Later work extended the application of reduced precision to inference post-training quantization [9], 8-bit matrix multiplication [8], activation stabilization [24], and activation-aware weight quantization [14]. These works proved that neural networks are generally robust to reduced precision, albeit focusing on training stability or model compression, not on the interaction between accumulation order and KV cache execution path. No prior paper isolates the order divergence caused systematically by FP16 non-associativity on deterministic cached inference.

B. KV CACHE OPTIMIZATION AND ATTENTION EFFICIENCY

FlashAttention [7] and its subsequent version [6] introduced IO-efficient exact-attention kernels that increase throughput by an order of magnitude while reducing memory access for transformer decoding; both implicitly rely on numerical equivalence across caching schemes. PagedAttention [13] offered scalable KV cache management for large-scale inference, explicitly optimizing memory use without analyzing differences in output correctness across execution paths. Multi-query attention [18] and grouped-query attention [2] reduce decoding costs by sharing keys and values across heads, but neither study considers the interaction between shared KV tensors and FP16 accumulation order, thus overlooking the effects on output self-consistency. This is the first work to systematically measure divergence caused by execution-path variations in KV caching and demonstrate that the GQA ratio can predict divergence magnitude through error propagation among queries amplified by FP16 accumulation.

C. MECHANISTIC INTERPRETABILITY

Causal tracing and ROME [15] introduced residual-stream interventions for discovering and editing facts in transformers, postulating the residual stream as the dominant site of causal intervention. Path patching [23] isolated causal pathways in residual-stream contributions for the indirect-object-identification task, while causal mediation analysis [22] formalized activation-level interventions for mechanistic attribution. Interpretable computational primitives such as induction heads were discovered under the assumption of a deterministic forward pass [17]. These methods all target the residual stream and implicitly assume that divergence originates there. Our activation patching yields a null result: residual-stream patching cannot recover KV cache divergence even when all layers are patched simultaneously. The KV cache state itself emerges as the causal variable, a cause neglected by existing work.

D. LLM OUTPUT CONSISTENCY AND REPRODUCIBILITY

Previous work has investigated variability in LLM output due to stochastic sampling, prompt sensitivity [28], dataset-specific limitations [3], or systemic reproducibility issues by comparing model versions and evaluation conditions [4], [1]. Those studies treat the source of output variability as orthogonal to the model itself and instead dependent on stochastic effects or distribution shift. Our work is fundamentally different: we elicit reproducible, deterministic divergence solely from the KV cache execution path under FP16 inference. This systematic reproducibility effect appears under greedy decoding and can be mechanistically explained by FP16 accumulation order, without the influence of any stochastic factor.

Most closely related to our work, Yuan et al. [26] show that FP16 and BF16 non-associativity can lead to reproducibility failures across system configurations such as batch size, GPU count, or hardware platform. They also introduce LayerCast, which stores weights in FP16 but computes in FP32, to mitigate this effect. Our findings are complementary but differ in three ways. First, we show divergence between cache-ON and cache-OFF paths on identical hardware with the same batch size, a deterministic effect unaffected by configuration. Second, we isolate the KV cache execution path as the source of divergence and localize it through activation patching, which Yuan et al. [26] do not examine. Third, we analyze how the GQA ratio serves as an indicator of divergence magnitude, which is not explored in that work but is highly relevant to contemporary LLM architectures.

APPENDIX B STATISTICAL ANALYSIS

Each test is selected to answer a particular inferential question in the argument rather than as a more general check of significance.

Behavioral significance. We apply McNemar’s test to the paired correctness results (cache-ON correct vs. cache-OFF correct on the same example). Pairing by example controls for prompt difficulty; the test asks specifically whether one execution path systematically outperforms the other, not merely whether accuracy differs. This is a secondary criterion for Step 1 because we need to verify that the direction of divergence is consistent rather than random. Multiple comparisons are corrected using Bonferroni and Benjamini–Hochberg (BH-FDR) at $\alpha = 0.05$; bootstrap confidence intervals ($B = 10,000$ resamples, 95% level) are given for mean KL and the difference in accuracy.

Layer drift significance. We run Mann–Whitney U tests on per-layer magnitudes of hidden-state drift across the 90 layer–model combinations, with BH-FDR correction. The nonparametric test is chosen because drift magnitudes are right-skewed and not normally distributed. This analysis supports Step 2 by indicating which layers reliably amplify divergence.

Step-count correlation. We use Pearson correlations between flip index and mean KL divergence ($n = 10,500$ per model). Because the flip-index distribution is heavily skewed

toward zero, we also report partial correlations excluding flip index = 1 and flip indices ≤ 3 . These correlations test whether the negative relationship is stronger without the most extreme boundary flips, providing a direct test of Step 4.

APPENDIX C EXPERIMENT ENVIRONMENT AND SUB-EXPERIMENTS

A. SYSTEM ENVIRONMENT

All experiments are conducted on a single NVIDIA H100 NVL (95 GB) GPU, paired with an AMD EPYC 9254 24-core CPU and 250 GB of RAM. The system runs Ubuntu 22.04 with Python 3.12.3, PyTorch 2.5.1 (CUDA 12.1), and Hugging Face Transformers 4.53.0.

For greedy decoding, we set `CUBLAS_WORKSPACE_CONFIG=:4096:8` from the attention architecture. The 0.036% average relative accumulation error is flat across sequence lengths and models, establishing that per-step error magnitude is determined by head dimension and attention structure, not input context. When combined with GQA, the effect becomes significantly worse because one FP16 error propagates to R query heads, producing R correlated errors rather than a single independent error term. Within-model layer-1 amplification (the ratio of layer-1 to layer-0 drift) reaches $105\times$ for LLaMA-2 and $35\times$ for Mistral, consistent with GQA simultaneously broadcasting the same error to all four query heads.

B. ROOT CAUSE SUB-EXPERIMENTS 1-4

Experiment 1 quantifies the relative FP16 accumulation error of dot products versus sequence length ($\approx 0.036\%$, flat across all lengths and models, standard deviation = 8.5×10^{-7}). Experiment 2 simulates multilayer error propagation through N sequential attention and residual steps, showing how single-layer FP16 rounding errors compound across layers, consistent with the large KL divergence observed in the behavioral characterization. Experiment 3 measures the FP16 rounding error in K and V projections between joint computation (cache-OFF) and token-by-token computation (cache-ON) to confirm that the error originates at cache-write time rather than downstream. Experiment 4 correlates the flip index with mean KL divergence to assess whether proximity to the decision boundary mediates the effect.

APPENDIX D EXTENDED MECHANISTIC DISCUSSION

Practical guidance for practitioners is given in Section VI; this appendix provides the deeper mechanistic account underlying those recommendations.

A. A UNIFIED MECHANISTIC ACCOUNT

The five experiments form one unifying argument. Behavioral characterization shows that cache-ON and cache-OFF always diverge in FP16 across every run, model, strategy, and seed. Root-cause falsification identifies the cause: FP16 non-associativity causes the two execution paths to accumulate in different orders, and FP32 eliminates the divergence. Layer-drift analysis demonstrates how the initial error propagates. The error first grows rapidly at layer 1 in both LLaMA-2 and Mistral and is broadly distributed across all layers in Gemma. Decision-boundary analysis shows when this matters behaviorally: not when the divergence is large in magnitude, but when the model is near a decision boundary. Causal localization establishes the component in which the error resides: patching the entire residual stream fails, even when all layers are patched simultaneously. The KV cache state carries the error forward.

This mechanism is deterministic, reproducible, architecturally predictable, and cannot be corrected by post-processing. It therefore differs clearly from the stochastic output fluctuations demonstrated in prior studies [28], [4], which depend on sampling temperature and prompt sensitivity, and from the batch-size- and hardware-dependent nondeterminism recently characterized by [26]. Our divergence requires neither: it appears on identical hardware with identical batch size, exists even under greedy decoding with a fixed seed, and can be eliminated by changing the arithmetic precision.

B. ARCHITECTURAL PREDICTABILITY

Divergence is not a random phenomenon; it can be predicted from the attention architecture. The 0.036% average relative accumulation error is flat across sequence lengths and models, establishing that per-step error magnitude is determined by head dimension and attention structure, not input context. When combined with GQA, the effect becomes significantly worse because one FP16 error propagates to R query heads, producing R correlated errors rather than a single independent error term. Within-model layer-1 amplification (the ratio of layer-1 to layer-0 drift) reaches $105\times$ for LLaMA-2 and $35\times$ for Mistral, consistent with GQA simultaneously broadcasting the same error to all four query heads.

We report layer-1 drift normalized by the cache-ON hidden state's own norm as primary (Mistral 902.36 exceeds LLaMA-2 361.96, consistent with the GQA-amplification account above), since raw (absolute) L2 magnitude is confounded by residual-stream scale: the raw ranking in fact reverses (LLaMA-2 922.29 exceeds Mistral 336.62), because residual-stream norm grows substantially and differently across models with depth (Appendix D-C). We flag this explicitly because it is a natural point of confusion: absolute magnitude comparisons across models are only meaningful once normalized by each model's own activation scale.

Gemma's 256-dimensional heads exhibit a clearly flat propagation pattern and broad, uniform accumulation rather than focused amplification, leading to the highest overall KL divergence despite the lowest per-layer drift. This behavior depends solely on the architecture and should be predictable before inference.

C. RAW VS. NORMALIZED LAYER-1 DRIFT

For transparency, Table 9 reports both quantities directly. The normalized ranking (Mistral > LLaMA-2) is consistent with the GQA-amplification account in Section III; the raw ranking reverses, illustrating that raw magnitude alone is not a reliable basis for cross-model comparison.

TABLE 9: Layer-1 drift: raw vs. normalized (mean and median across examples).

Model	Raw L2 (mean)	Normalized (mean)	Normalized (median)
LLaMA-2-7B	922.29	361.96	360.62
Mistral-7B	336.62	902.36	897.34
Gemma-2-2B	74.09	1.00	1.03

D. THE LOCALIZATION ARGUMENT AND ITS LIMITS

Causal localization establishes that divergence does not reside in the residual stream; direct patching of the KV cache tensors (Section V-F) converts this elimination argument into positive confirmation, subject to the scope noted there (joint K/V patching, step-0 measurement only, single direction tested). The current results together establish a necessary condition (no residual-stream intervention recovers the divergence) and, within that scope, a sufficient one at decode step 0 (restoring the KV cache state does).

E. IMPLICATIONS

(i) Cache equivalence cannot be assumed. All FP16 production systems leveraging KV caching yield results distinct from those of no-cache inference—in 100% of cases. Validation pipelines that compare cached and noncached results will report these differences as properties intrinsic to FP16 rather than as errors. **(ii) Divergence direction is systematic.** Under our evaluation protocol, cache-ON yields higher accuracy than cache-OFF in 8 of 9 conditions. Because an execution-path difference that merely introduced an unbiased perturbation would be expected to help and hurt accuracy in roughly equal measure across conditions, this consistent directionality indicates systematic execution-path bias rather than symmetric noise. **(iii) GQA amplifies the effect.** Such divergence will be even more common with the advent of GQA and MQA, which are increasingly dominant as modern LLMs become larger. **(iv) The effect is not specific to FP16.** BF16 reproduces the divergence at statistically indistinguishable severity from FP16, consistent with a mantissa-precision mechanism common to both formats rather than one specific to FP16's narrower exponent range. Practical guidance on detection, mitigation trade-offs, and architecture-specific considerations arising from these implications is given in Section VI.

F. LIMITATIONS

(i) Scope. Three models and one benchmark represent important architectural dimensions, though not the full spectrum in use. We partially extend the scope along two axes: we confirm that 100% token divergence persists at `max_new_tokens=400` (vs. 128) across all three base models and, separately, on Mistral-7B-Instruct-v0.3 using its native chat template. These results indicate that the effect is neither an artifact of truncation length nor specific to base (non-instruction-tuned) models. Full validation at generation lengths typical of extended reasoning traces (e.g., AIME- or LiveCodeBench-style outputs, potentially several thousand tokens) on reasoning-tuned models and on other benchmarks and hardware platforms remains future work. Our root-cause analysis (per-step accumulation error determined by head dimension, not sequence length or content) predicts that the effect should persist or compound at such lengths, but this prediction has not yet been directly confirmed at that scale. Separately, the sequence-length-independence claim is limited to the tested range of 16–128 tokens. The per-step accumulation error is determined by head dimension

rather than sequence length, so the 0.036% figure is expected to remain stable per step at longer contexts. However, in long-context settings (4K+ tokens), each decode step writes a new error-contaminated KV entry that persists in the cache for all future steps; the cumulative divergence effect is therefore expected to grow monotonically with context length, making long-context settings a priority for future experimental characterization. **(ii) Evaluation protocol.** Under the current protocol, the absolute accuracy values are low because of truncation at `max_new_tokens=128` and the few-shot prompt format. Moreover, accuracy differences serve as indicators of execution-path bias under these conditions, not as measures of reasoning performance. The core behavioral results—100% divergence across all conditions and a systematic direction in 8 of 9 conditions—do not depend on the level of absolute accuracy. We additionally verify that token-level divergence corresponds to practically meaningful outcome differences: 98.4% of divergent examples produce different final answers, exceeding the reconvergence failure rate under ordinary sampling variance alone (Section IV-C). **(iii) Direct KV-cache patching: scope, not absence.** Direct patching of KV cache tensors (Section V-F) converts the residual-stream elimination argument into positive confirmation, but three specific extensions remain open: the intervention patches K and V jointly rather than separately, so the individual contribution of each is not isolated; the patch is applied and measured only at decode step 0, so persistence across a full generated trajectory under repeated patching is untested; and only the cache-ON-into-cache-OFF direction was run, leaving a symmetric reverse-direction check (patching cache-OFF into cache-ON) as an additional robustness test for future work. **(iv) Quantized inference.** This paper studies FP16 inference; integer-quantized models (INT8, INT4/GPTQ, AWQ) represent a distinct regime that is beyond our scope. When KV tensors are quantized to INT8 before being written to cache, the dominant source of numerical error shifts from FP16 rounding (approximately 0.036% relative error, as measured here) to INT8 quantization noise, which is larger in magnitude but applies symmetrically across both cache-ON and cache-OFF paths if the quantization grid is identical. Our mechanism requires the exact FP16 mantissa error to be stateful across decode steps; under INT8 KV quantization, this FP16 error would be dominated by the quantization noise floor and would not be separately observable. Whether INT8 KV caching introduces its own form of cache-ON/cache-OFF inequivalence through asymmetric quantization rounding between jointly computed and token-by-token computed keys and values is a separate question that we identify as a natural extension of this work.

APPENDIX E FUTURE WORK

The most proximal remaining extensions to the direct KV cache tensor patching result (Section V-F) are: separating the causal contribution of K versus V tensors rather than patching them jointly; testing whether patching at every decode step (rather than step 0 only) sustains trajectory-level agreement

with cache-ON across a full generation; and running the symmetric reverse-direction patch (cache-OFF into cache-ON) as an additional robustness check. None of these are blocked on tooling – the current implementation already supports arbitrary layer/step targeting – and each is a natural next increment on the existing pipeline.

We partially addressed the question of mitigation beyond full FP32: a mixed-precision ablation confining elevated precision to the attention computation alone showed no measurable benefit, narrowing (but not closing) the search space. Other mechanisms remain worth investigating: periodic cache refresh (recomputing KV tensors from scratch at fixed intervals) and smoothed KV updates (averaging cached and recomputed tensors to reduce drift) were not tested here and trade compute or memory for reduced divergence, potentially more cheaply than full FP32 inference in production.

We confirmed that the divergence persists under BF16 at a severity statistically indistinguishable from that under FP16 and separately confirmed 100% token divergence at longer generation lengths (400 tokens) and on an instruction-tuned model (Mistral-7B-Instruct-v0.3). Open questions include formal quantification of the GQA amplification factor across a wider range of sharing ratios with matched parameter counts (isolating attention grouping as the sole variable rather than the covarying architectural differences across our three tested models); validation at reasoning-model-scale generation lengths (potentially several thousand tokens, as in AIME- or LiveCodeBench-style traces); and characterization of the interplay between KV cache divergence and cache-compression schemes (e.g., quantization, eviction, and low-rank methods), for which additional precision loss may compound the effects reported here.

Decision-boundary analysis shows that behavioral consequences depend on proximity to the decision boundary but does not specify how much accumulated divergence is required to push a state across a boundary. Adversarial-robustness-style experiments that inject prespecified noise directly into KV tensors and measure flip rate as a function of noise amplitude would determine susceptibility thresholds and clarify how close to a decision boundary a model must be for divergence of a given magnitude to produce an observed behavioral flip. Directly instrumenting the cache runtime to track how FP16 rounding errors accumulate in K and V tensors during decoding, rather than inferring accumulation indirectly from output distributions, would similarly relate the accumulation rate and behavioral disparity more precisely.

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